SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
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ITA0329 - MOBILE COMPUTING

ASSIGNMENT 1 – CO1

CO1 Coverage: Cellular network concepts; Frequency reuse and interference in cells; Access channels and mobile network capacity; Fundamentals of mobile computing

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Assignment Title: Design, Analyse and Optimise a Hexagonal Cellular Network for Spectrum Allocation, Interference Management and Scalable Coverage Planning in a University Township (Total Marks – 50)

Duration: 1week

Scenario — Cellular Coverage Planning for a University Township

An operator has been licensed to provide cellular service to a 12 km² university township containing academic blocks, hostels, residential colonies and a main arterial road. The operator holds a fixed allocation equivalent to 336 duplex voice channels (after control-channel overhead) and must plan a hexagonal-cell network. Subscriber demand will grow as the township expands.

You are the radio-planning team. You must investigate how spectrum can be shared across cells, how interference limits that sharing, and how the design must change as the subscriber base grows.

Planning data and constraints

- Subscribers: 1,000 → 10,000 → 20,000 → 100,000+ (planning horizon).
- Average busy-hour traffic per subscriber: 0.025 Erlang. Target grade of service: ≤ 2% call blocking.

- Minimum acceptable SIR: choose and justify a threshold between 12 dB (digital-style) and 18 dB (analogue-style).
- Path-loss exponent n: 2.5 (open campus) to 4 (dense built-up area); justify the values you adopt.
- Spectrum cannot be increased; only the network layout and radio parameters may change.
- Any planning effort (surveys, traffic estimation, number of sites) must be included in your cost analysis.

Design Challenge (Day 4):

A convocation / sports event will bring 25,000 attendees (0.05 Erlang each) into a 1 km² stadium zone for four hours. Re-evaluate your original design and decide whether to retain, modify or redesign it.

TASK 1 – Design Alternative Frequency-Planning Strategies 5 Marks

Guiding Enquiry

How can a fixed block of spectrum be shared across many cells so that every subscriber is served without intolerable co-channel interference — and in how many fundamentally different ways can this be done?

Investigate and interpret:

- Propose at least **three meaningfully different strategies** (for example fixed cluster reuse, sectorised reuse, dynamic/borrowed channel assignment, hierarchical macro–micro layering). The same strategy with different cluster sizes (N = 3 vs N = 7) does not count as two strategies.
- At least one strategy must be expressed as an **iterative procedure** (e.g., searching valid cluster sizes $N = i^2 + ij + j^2$ and assigning channel sets across the grid) and at least one as a **recursive/hierarchical procedure** (e.g., repeated cell splitting until demand is met).
- analyse the basic idea, the assumptions, and the way the **access channels** are shared (FDMA/TDMA/CDMA/OFDMA concept and control vs traffic channels).
- Provide pseudocode or a flowchart for each, name the design principle used (spatial reuse, trunking, spatial filtering), and justify why each deserves investigation.
- State the preprocessing required (traffic map, propagation survey, hex-grid layout) and who bears that cost.

Evidence Expected

Three distinct strategies with pseudocode/flowcharts, identified design principle, preprocessing requirements and justification for each.

Answer:

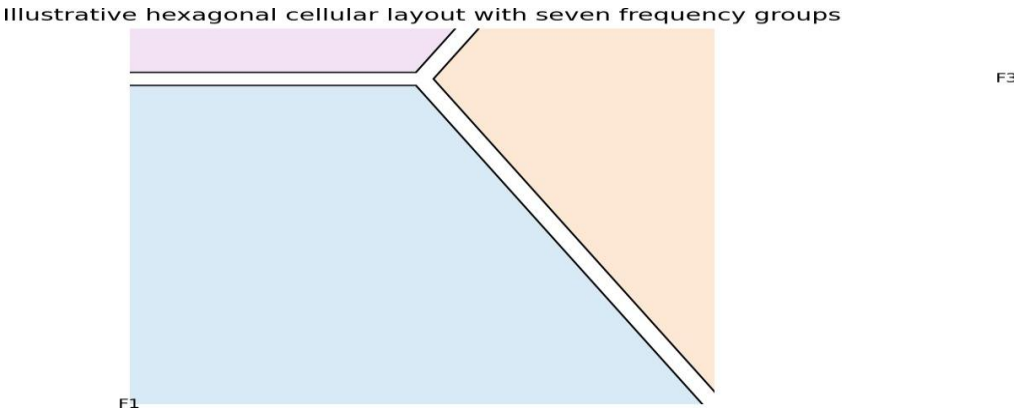
The central engineering problem is to reuse a fixed spectrum allocation across many cells while keeping co-channel interference below the selected SIR threshold. A frequency-planning strategy defines how channels are distributed spatially, how interference is controlled and how capacity is increased when traffic grows.

1.1 Strategy A: Fixed Cluster Frequency Reuse

In fixed cluster reuse, the 336 duplex traffic channels are divided into N disjoint channel groups. Each cell receives one group, and the same group is reused only in cells separated by a prescribed reuse distance. The cluster is repeated over the hexagonal grid. The cluster size must be a valid integer of the form $N = i^2 + ij + j^2$, where i and j are non-negative integers.

- Access concept: in a classic FDMA system, a channel is a frequency pair; in TDMA, each carrier is divided into time slots. The same spatial reuse principle applies to a carrier/time-slot resource block.
- Control channels carry registration, paging, access requests and handoff signalling. Traffic channels carry voice or user payload. The assignment states 336 channels after control-channel overhead, so the calculations treat all 336 as usable traffic channels.
- Design principle: spatial reuse. A frequency group is reused in sufficiently distant cells to multiply the geographical use of the fixed spectrum.

Assumptions: approximately regular hexagonal cells; a planned reuse pattern; similar channel demand between cells; stable propagation conditions during the planning interval; and a network-wide channel plan. The design is simple and predictable but can be inefficient when traffic is uneven.

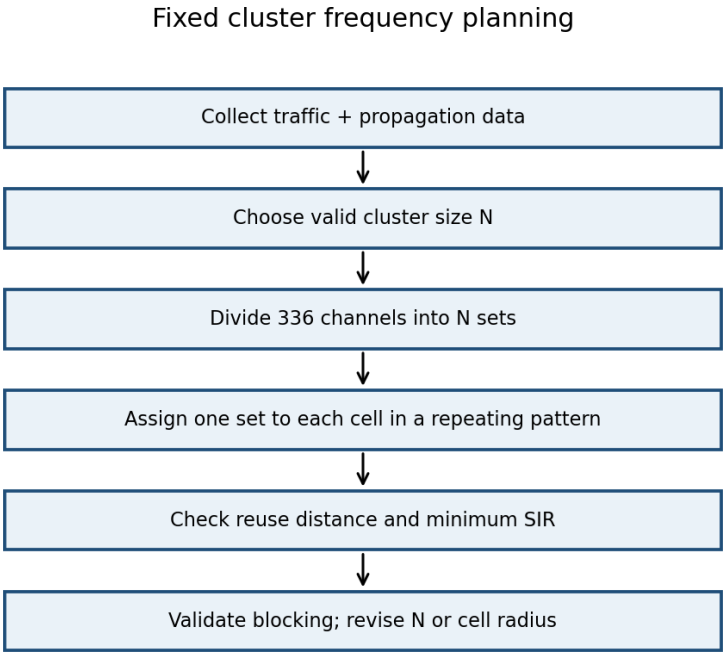


1.2 Fixed reuse iterative procedure

The iterative procedure searches candidate (i,j) pairs, computes N, divides the channel pool and evaluates interference. It rejects a cluster size if the calculated SIR is below the target or if the per-cell channel pool cannot meet the offered traffic at 2% blocking.

INPUT: C = 336 channels, R, path-loss exponent n, SIR_min
valid_pairs = [(i,j) for i in 0..m for j in 0..m]

```
FOR each (i,j):
    N = i*i + i*j + j*j
    IF N == 0: continue
    channels_per_cell = floor(C/N)
    Q = sqrt(3*N)
    D = Q*R
    SIR = ( (D/R)**n ) / 6      # six equal first-tier interferers
    IF SIR >= 10**(SIR_min/10):
        evaluate ErlangB(channels_per_cell, 0.02)
        retain feasible candidate
SELECT candidate balancing capacity, SIR and site cost
ASSIGN N channel groups and repeat cluster over grid
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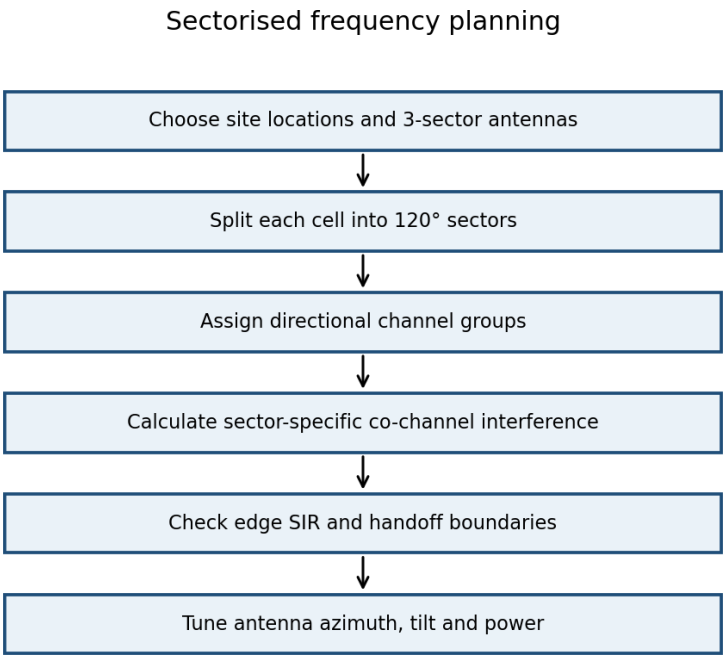


1.3 Strategy B: Sectorised frequency reuse

Sectorisation divides a cell site into directional coverage sectors, commonly three 120° sectors. Each sector uses a directional antenna rather than an omnidirectional antenna. The antenna gain concentrates transmitted and received energy toward the intended sector and attenuates signals arriving from other directions. This reduces the number of strong first-tier co-channel interferers seen by a mobile in a sector.

- Access concept: FDMA/TDMA carriers or OFDMA resource blocks can be assigned per sector. Control signalling remains available in each sector or is coordinated at site level.
- Design principle: spatial filtering. Directional antenna patterns suppress unwanted signals and permit a tighter reuse plan when measured SIR allows it.
- Assumptions: antenna azimuth and downtilt are correctly engineered; sector boundaries are surveyed; side-lobe and back-lobe levels are included in realistic simulation; handoff between sectors is supported.

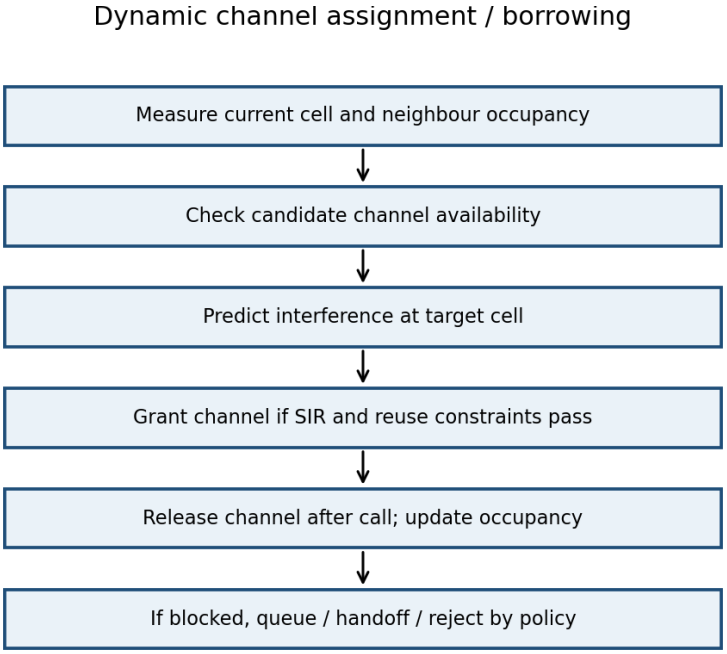
For a simplified three-sector model, the dominant co-channel interferer count may fall from six to approximately two in the relevant sector. This is an idealised first-tier approximation, not a universal guarantee. Real antenna patterns and sector orientation determine the actual interference.



1.4 Strategy C: Dynamic channel assignment and borrowing

Dynamic channel assignment (DCA) selects a channel at call setup from a shared pool after checking occupancy and interference constraints. Channel borrowing is a related approach: a congested cell temporarily uses an otherwise idle channel from a neighbour, provided the reuse constraint remains satisfied. This adapts to non-uniform demand such as lecture halls, hostels and the stadium.

- Access concept: a channel may be a frequency carrier, a carrier/time-slot pair, or an OFDMA resource block. The controller tracks active assignments and control-channel state.
- Design principle: trunking efficiency and interference-aware resource allocation. A common or borrowable pool reduces the chance that one cell blocks calls while a neighbour has idle channels.
- Assumptions: a centralised or distributed controller has timely occupancy data; neighbour relations are correct; channel assignment latency is acceptable; and interference checks use measured or predicted propagation.

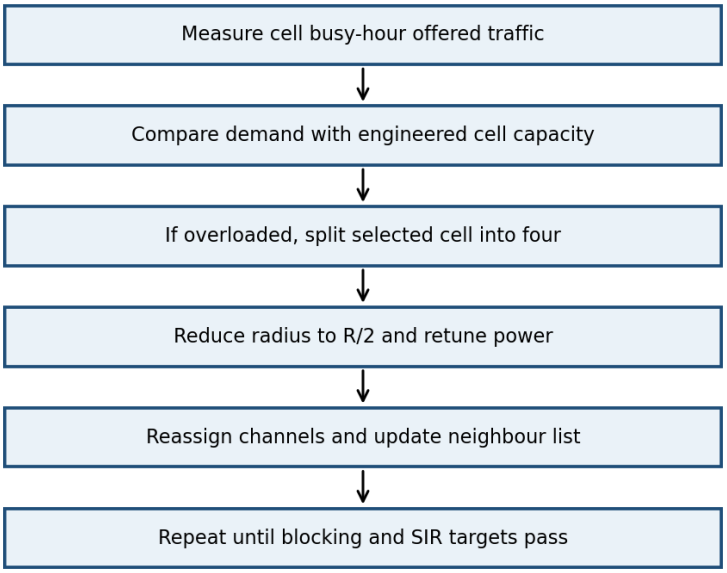


1.5 Strategy D: Hierarchical macro–micro layering and cell splitting

A hierarchical network uses macro cells for broad-area mobility and coverage, while microcells or picocells provide extra capacity in hotspots. Repeated cell splitting is a recursive form of densification: an overloaded cell is replaced by smaller cells, each with a smaller radius and a new local frequency plan. This is distinct from merely choosing a different cluster size.

- Access concept: the same radio access technology may be used across layers, with coordinated frequency/resource allocation. Macro layer handles mobility and coverage; micro layer absorbs concentrated traffic.
- Design principle: spatial densification and load offload. More radio sites are introduced where demand is high instead of increasing the licensed spectrum.
- Assumptions: backhaul and power are available at added sites; handoff parameters are tuned; interference coordination exists between layers; and site acquisition is possible.

Hierarchical cell-splitting procedure



1.6 Preprocessing, ownership and justification

Preprocessing item	What must be prepared	Who bears the cost
Traffic map	Busy-hour subscriber density by academic blocks, hostels, residences, roads and stadium	Operator radio-planning team; university supplies anonymised occupancy schedule
Propagation survey	Received signal, path loss, building penetration, shadowing and interference measurements	Operator; specialist drive-test / walk-test contractor
Hex-grid layout	Candidate site coordinates, cell radius, neighbour relations and boundary coverage	Operator planning and optimisation team
Channel inventory	Carrier/channel list, control-channel reservation and equipment compatibility	Operator spectrum and RAN engineering teams
Capacity forecast	Subscriber growth, Erlang traffic, event calendar and 2% blocking target	Operator with university administration
Cost model	Survey hours, site rental, radio equipment, backhaul, power and maintenance	Operator finance and deployment teams

Why investigate all strategies? Fixed reuse gives a transparent and low-complexity baseline. Sectorisation improves interference management without requiring new spectrum. Dynamic assignment improves statistical multiplexing when demand is uneven. Hierarchical layering and splitting address concentrated and long-term growth, but introduce site, backhaul, handoff and coordination costs.

TASK 1 – Strategy comparison and design selection logic

Strategy	Main resource-sharing method	Strength	Limitation	Best use case
Fixed cluster reuse	Preassigned frequency groups repeated every N cells	Predictable, easy to audit and deploy	Uneven demand wastes idle channels; cluster size trades capacity for SIR	Uniform township-wide baseline
Sectorised reuse	Directional sectors with spatially separated reuse	Fewer dominant interferers; improved SIR	Antenna complexity, sector-edge handoffs, side-lobes	Dense blocks and high-demand corridors
Dynamic assignment / borrowing	Channels selected from pool based on occupancy and interference	Adapts to hotspots; better trunking efficiency	Control complexity; borrowing can create interference elsewhere	Variable lecture/hostel/event traffic
Macro–micro layering / splitting	Add smaller cells under macro coverage	High local capacity and targeted densification	More sites, handoffs, backhaul and planning cost	Stadium and long-term growth hotspots

1.7 Channel-sharing concepts: FDMA, TDMA, CDMA and OFDMA

FDMA divides spectrum into frequency channels. A call is assigned a frequency pair, and co-channel cells reuse the same frequency only after adequate spatial separation. TDMA divides a carrier into repeating time slots; a user receives a time slot on a carrier. In FDMA/TDMA systems, the frequency plan and time-slot allocation are jointly considered.

1.8 Task 1 conclusion

The recommended investigation set is fixed N=7 reuse as the reference, three-sector reuse as an interference-reduction alternative, and dynamic channel assignment as the traffic-adaptive alternative. Macro–micro layering and cell splitting are retained as the growth mechanism. The strategies are meaningfully different because they use spatial reuse, directional spatial filtering.

TASK 2 – Identify the Governing Design Parameter 2 Marks

Guiding Enquiry

For each strategy, which single quantity most strongly controls capacity and interference, and why should the analysis be built around it?

Investigate and interpret:

- Identify the dominant parameter(s) for each strategy (e.g., cluster size N, reuse ratio Q = D/R, number of co-channel interferers, cell radius R, channels per cell).
- analyse where it appears in the strategy and why it is appropriate for efficiency analysis.
- interpret how this quantity must change as subscribers grow from 1,000 to 100,000+.

Evidence Expected

Correct identification and justification of the governing parameter for each strategy, linked to its behaviour.

Answer:

A governing parameter is the quantity that most directly controls the trade-off under investigation. It is not necessarily the only parameter; it is the first parameter around which the design analysis is organised.

Strategy	**Dominant parameter**	**Where it appears**	**Why it governs design**	**Growth response**
Fixed cluster reuse	Cluster size N (or reuse ratio $Q=\sqrt{3N}$)	Channel division C/N and reuse distance $D=R\sqrt{3N}$	Increasing N improves co-channel separation but reduces channels per cell	At higher demand, reduce N only if SIR permits; otherwise add cells/sectoring
Sectorised reuse	Effective co-channel interferer count I and antenna discrimination	SIR denominator and sector antenna pattern	Fewer strong interferers can permit tighter reuse while retaining SIR	Increase sectorisation/antenna coordination in dense areas
Dynamic assignment / borrowing	Available-channel occupancy /	Admission and borrowing decision at call setup	Determines whether idle capacity	Larger pool and smarter allocation become more
	blocking state per cell		elsewhere can serve a hotspot	valuable under uneven demand

Hierarchical macro–micro	Cell radius R (or site density $1/R^2$)	Coverage area, cells per km^2, path loss and handoff rate	Smaller R increases spatial capacity but also sites and handoffs	Reduce R selectively in hotspots

2.1 Why cluster size N is central to fixed reuse

The 336 channels are divided among N cells in a cluster. The approximate channels per cell are S=C/N. Thus, larger N means fewer channels available in each cell. At the same time, the reuse distance D increases as $\sqrt{N}$, so co-channel interference falls. N therefore controls the fundamental capacity–interference trade-off.

For example, N=7 gives 336/7=48 channels per cell. N=4 gives 84 channels per cell but a smaller reuse distance and a lower idealised SIR. N=12 gives 28 channels per cell and better isolation. This is why the design cannot maximise capacity by choosing the smallest N without checking SIR.

2.2 Why R governs densification

The hexagonal cell area is proportional to R^2. Halving R reduces the area of one cell to one quarter. The number of cells needed for a fixed geographical area therefore increases by approximately four times. This raises the number of sites and handoff boundaries but also makes the network capable of serving more local traffic.

2.3 Parameter evolution from 1,000 to 100,000+ subscribers

Subscribers	Busy-hour offered traffic	Engineering implication
1,000	25 Erlangs	Sparse initial demand; a small number of sites may be sufficient, subject to coverage
10,000	250 Erlangs	Capacity begins to require careful channel allocation and traffic balancing
20,000	500 Erlangs	Hotspot analysis, sectorisation and selective densification become important
100,000	2,500 Erlangs	A single macro-only layer is unlikely to be economical; multiple layers and extensive reuse planning are required

The total offered traffic follows A=0.025U, where U is the number of subscribers. If the subscriber population increases by a factor of 100, the offered traffic also increases by a factor of 100. The spectrum does not increase, so capacity growth must come from spatial reuse, sectorisation, improved trunking efficiency and smaller cells.

TASK 3 – Derive the Capacity and Interference Functions Mathematically 6 Marks

Guiding Enquiry

Starting from hexagonal geometry, how do coverage, capacity and interference depend on the design choices you made?

Investigate and interpret:

- Derive the cluster-size relation, reuse distance $D = R\sqrt{(3N)}$ and reuse ratio Q, hexagonal cell area, and the number of cells needed to cover 12 km².
- Derive channels per cell (S/N), traffic per cell at 2% blocking (Erlang-B reasoning), and total system capacity as a function of N and R.
- Derive the co-channel SIR expression for each strategy (first-tier interferers; effect of sectoring on the number of interferers).
- Combine **planning cost** (survey, number of sites) with **operating capacity** to obtain an overall cost–capacity relation for each strategy.
- Show every step; stating a final formula without derivation will not earn full credit.

Evidence Expected

Clear derivations of capacity, number of cells and SIR for each strategy, including planning cost and overall cost–capacity relation.

Answer:

3.1 Hexagonal cluster-size relation

A regular hexagonal cell has six neighbours. To reach an equivalent cell in the next co-channel cluster, move i cell steps along one chain of hexagons, turn 60 degrees and move j steps. The squared length of this displacement, measured in units of cell-centre spacing, is $i^2 + ij + j^2$. Therefore the cluster size is:

$N = i^2 + ij + j^2$

Examples of valid cluster sizes include N=1 (i=1,j=0), N=3 (1,1), N=4 (2,0), N=7 (2,1), N=9 (3,0), N=12 (2,2), N=13 (3,1), N=16 (4,0). The assignment specifically requests tests N=3,4,7,12. N=7 is selected for the baseline because it is a common reuse pattern and satisfies the 18 dB target in the n=4 six-interferer approximation.

3.2 Reuse distance and reuse ratio

Let R be the cell radius (centre to vertex). The distance between adjacent cell centres is $\sqrt{3}R$. The co-channel reuse distance is:

$D = \sqrt{3}R \times \sqrt{N} = R\sqrt{(3N)}$

The reuse ratio is the reuse distance divided by cell radius:

$Q = D/R = \sqrt{(3N)}$

N	Valid (i,j)	Q=√(3N)	D when R=0.5 km
3	(1,1)	3.000	1.500 km
4	(2,0)	3.464	1.732 km
7	(2,1)	4.583	2.291 km
12	(2,2)	6.000	3.000 km

3.3 Hexagonal cell area and number of cells

A regular hexagon can be divided into six equilateral triangles of side R. Area of one equilateral triangle is $(\sqrt{3}/4)R^2$. Therefore:

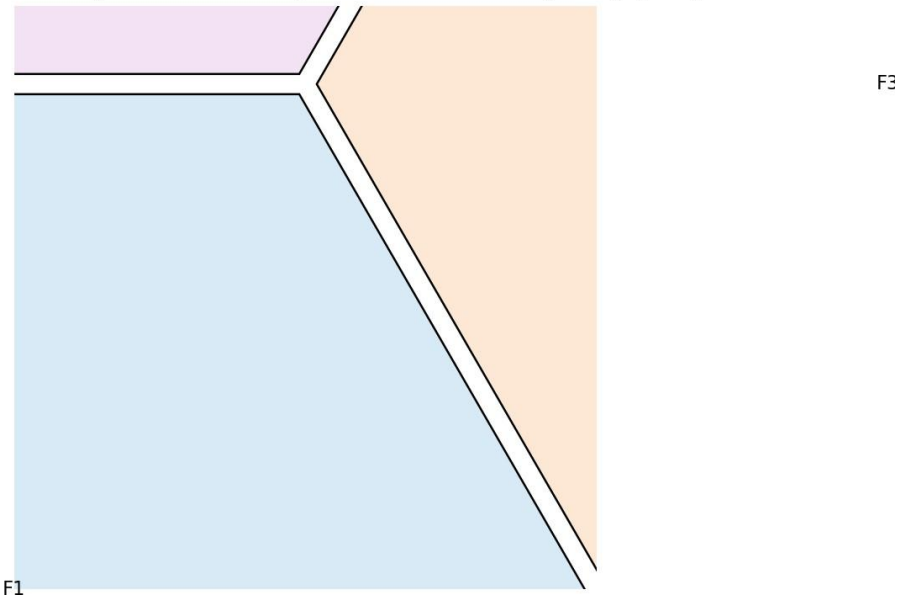
$A_{cell} = 6 \times (\sqrt{3}/4)R^2 = (3\sqrt{3}/2)R^2 \approx 2.598R^2$

For R=0.5 km, $A_{cell} = 2.598 \times 0.25 = 0.6495$ km². The ideal area-only cell count for A_{town}=12 km² is:

$M = \text{ceil}(A_{town}/A_{cell}) = \text{ceil}(12/0.6495) = \text{ceil}(18.475) = 19$ cells.

This is a first-order estimate. A real network needs edge coverage, overlap, indoor penetration and road corridors; the final site count may be higher. The count is not the same as a guarantee that 19 sites provide 100% coverage.

Illustrative hexagonal cellular layout with seven frequency groups



3.4 Channel allocation per cell

The total traffic-channel pool is C=336. Under fixed reuse, the channels are divided equally among N frequency groups. The nominal number of channels per cell is:

$S = C/N$

N	C/N	Integer channels per cell	Comment
3	112	112	High channel count per cell; tighter reuse and more interference risk
4	84	84	High nominal capacity; SIR must be verified
7	48	48	Baseline compromise
12	28	28	More reuse separation but fewer channels per cell

The calculation assumes equal channel groups and ignores guard-channel asymmetry or reserved channels beyond the stated control-channel overhead. If the division is not an integer, practical channel sets must be allocated as evenly as possible, with the remainder distributed among selected cells.

3.5 Offered traffic and Erlang-B blocking

One Erlang represents one channel occupied continuously for one hour. The busy-hour offered traffic from U subscribers is $A=U\times0.025$ Erlangs. A cell with S traffic channels is modelled as an M/M/S/S loss system: there are S servers, no waiting room, and an arriving call is blocked if all channels are busy.

The Erlang-B recursion is:

$B(0,A)=1;\ B(c,A) = [A\ B(c-1,A)] / [c + A\ B(c-1,A)]$

For a fixed number of channels c, the largest offered load A_c satisfying $B(c,A_c) \leq 0.02$ is the engineered traffic capacity per cell. Erlang-B is preferable to simply assuming one channel equals one Erlang because it explicitly accounts for call blocking and statistical multiplexing.

Channels c	Max offered load at 2% blocking (Erlangs)	Blocking at rounded load
28	20.15	0.0200
48	38.39	0.0200
84	72.53	0.0200
112	99.62	0.0200

For the baseline N=7, each cell has 48 channels and supports approximately 38.39 Erlangs at 2% blocking under the Erlang-B model. With 19 cells, the theoretical aggregate carried offered-load limit is approximately 729.44 Erlangs, assuming balanced traffic and no additional signalling or outage constraints.

3.6 Total capacity as a function of N and R

The number of cells is approximately M(R)=ceil[A_town/(2.598R²)]. The number of channels per cell is S(N)=floor(C/N). The total offered traffic supported at the target blocking probability is:

$A_{total}(N,R) \approx M(R) \times A_{EB}(S(N), 0.02)$

The subscriber capacity is:

$U_{max}(N,R) \approx A_{total}(N,R) / 0.025$

This expression is an idealised balanced-load upper estimate. It assumes each cell has comparable traffic and that each cell can use its full Erlang-B capacity. If demand is concentrated in one building or stadium, the hotspot may block even while other cells have idle channels. DCA and sectorisation address this mismatch.

3.7 Co-channel SIR for omnidirectional reuse

The received power from a transmitter at distance d is modelled as proportional to $d^{(-n)}$, where n is the path-loss exponent. Let the desired base station be at distance r from the mobile and let the first-tier co-channel interferers be at distances $d_1,\ldots,d_i$. Ignoring thermal noise for an interference-limited approximation:

$SIR = (P_0 r^{(-n)}) / \Sigma(P_i d_i^{(-n)})$

If all base stations transmit with equal power and the mobile is near the cell edge, the desired distance is approximated by $r \approx R$ and the six first-tier co-channel interferers are approximated as equidistant at D. Then:

$SIR \approx R^{(-n)} / [6D^{(-n)}] = (D/R)^n/6 = Q^n/6$

In decibels: $SIR_{dB} = 10 \log_{10}(Q^n/6)$. For N=7, $Q=\sqrt{21}=4.5826$. For n=4:

$SIR = 4.5826^4/6 = 441/6 = 73.5$; $SIR_{dB} = 10\log_{10}(73.5) \approx 18.66$ dB.

This reproduces the assignment sanity check of approximately 18.7 dB.

N	Q	SIR dB at n=2.5	SIR dB at n=3	SIR dB at n=4
3	3.000	4.15	6.53	11.30
4	3.464	5.71	8.41	13.80
7	4.583	8.75	12.05	18.66
12	6.000	11.67	15.56	23.34

3.8 Sectorised SIR approximation

For a sectorised cell, the antenna pattern reduces interference from many directions. In a simplified 120° sector model, assume only I=2 dominant co-channel interferers contribute strongly. If the interferers are treated as equidistant at D and the desired signal remains at R:

SIR_sector ≈ Q^n/I = Q^n/2

For N=7 and n=4, SIR_sector ≈ 441/2 = 220.5, or 23.43 dB. This simplified result is 4.77 dB higher than the omnidirectional six-interferer result because $10\log_{10}(6/2)=4.77$ dB.

This is an idealised bound. A realistic calculation must include antenna gain in the desired direction, side-lobe attenuation, back-lobe response, sector azimuth, tilt and the actual distances to interferers. A sector-edge mobile may see more than two significant interferers.

3.9 SIR for dynamic channel assignment

DCA does not have one fixed SIR formula because the selected channel depends on the active channel map. At call setup, a candidate channel f is accepted only if the predicted or measured SIR for all affected links meets the threshold. The general condition is:

$SIR_f = P_{desired} / (\Sigma P_{cochannel} + N_0) \geq \gamma_{min}$

The algorithm can use a reuse-exclusion list, received-signal measurements or an interference graph. Borrowing a channel from a neighbour is allowed only if the donor cell and all co-channel reuse cells remain above γ_{min}. The governing quantity is therefore the minimum predicted SIR over the set of affected mobiles, not merely the nominal cluster size.

3.10 Planning and operating cost model

The assignment explicitly requires survey effort and number of sites to be included. Since no rupee prices are supplied, define cost variables rather than inventing market prices:

Symbol	Meaning	Unit / interpretation
C_survey	Radio survey and traffic-study cost	One-time project cost
c_site	Site acquisition / rental / civil cost per site	Currency per site
c_radio	Radio equipment and installation cost per site	Currency per site
c_bh	Backhaul and power provisioning per site	Currency per site
c_opt	Annual optimisation and maintenance cost per site	Currency per site-year
M(R)	Number of required sites	Sites
A_cap	Total offered traffic capacity at target GoS	Erlangs

For a fixed-reuse design, an illustrative deployment cost is:

$C_{fixed} = C_{survey} + M(R)(c_{site} + c_{radio} + c_{bh}) + T \cdot M(R) c_{opt}$

where T is the operating period in years. Cost per supported Erlang is C_fixed/A_cap. Cost per supported subscriber is C_fixed/(A_cap/0.025). This allows a fair comparison when actual vendor and site quotations become available.

3.11 Strategy-specific cost–capacity relation

Strategy	Capacity expression	Cost expression / extra cost driver
Fixed reuse	M(R)·A_EB(floor(C/N),0.02)	C_survey + M(R)·(site + radio + backhaul) + maintenance
Sectorised reuse	M(R)·A_EB(S_sector,0.02), subject to sector SIR	Fixed cost + sector antennas, feeders, installation and optimisation
DCA / borrowing	Capacity from pooled/unequal cell loads with interference admission	Fixed cost + controller/software, measurement and signalling overhead
Macro–micro	Macro capacity + Σ microcell capacities	Macro cost + hotspot micro-sites + backhaul + handoff optimisation

DCA may improve carried traffic without increasing the number of channels because idle channels in low-demand cells can be used by busy cells. Sectorisation may permit a smaller reuse cluster, increasing channels per sector, but only if measured SIR remains acceptable. Cell splitting increases site count approximately fourfold for each uniform halving of radius.

3.12 Worked baseline interpretation

Baseline assumptions: N=7, R=0.5 km, n=3 for a mixed campus, and 19 ideal area-covering cells. Channels per cell = 48. The SIR at n=3 is $10\log_{10}(21^{(1.5)}/6) \approx 13.43$ dB, which does not meet the conservative 18 dB threshold. At n=4 the same idealised SIR is 18.66 dB. This shows why the baseline must not be approved using only one path-loss value: the mixed-campus n=3 case needs sectorisation, a larger reuse cluster, or measured evidence that the actual geometry/antenna pattern provides sufficient SIR.

If the strict 18 dB target is applied under the six-equal-interferer approximation, N=7 is acceptable at n=4 but not at n=2.5 or n=3. N=12 improves SIR but reduces channels per cell to 28. A practical plan can use a conservative cluster in open or weakly attenuating areas and sectorised/dynamic allocation in dense hotspots, provided the interference plan is validated end-to-end.

TASK 4 – Analyse Best, Average and Worst-Case Behaviour **4 Marks**

Guiding Enquiry

Where in the cell, and at what time of day, does each strategy perform best, typically, and worst — and why?

Investigate and interpret:

- Identify the position of the mobile (near the base station, mid-cell, cell corner) that yields best, average and worst SIR, and the geometry of interferers in each case.
- Identify the traffic condition (off-peak, average, busy-hour, event surge) that yields each blocking behaviour.
- analyse how shadowing and a changing path-loss exponent shift each case.
- Support each case with calculations, not just a table of values.

Evidence Expected

Justified best/average/worst analysis for SIR and blocking, supported by mathematical reasoning.

Answer:

4.1 Spatial positions and SIR

The best SIR normally occurs near the serving base station and in a direction where the serving antenna has high gain while co-channel interferers are distant or attenuated. The average case occurs at representative user locations distributed across the cell. The worst case often occurs near a cell corner or sector boundary, where the desired signal is weaker and two or more neighbouring co-channel signals may be relatively strong.

Case	Mobile position	Desired/interference geometry	Expected SIR behaviour
Best	Near serving base station	Small desired-link distance; interferers remain relatively far	Highest SIR
Average	Typical random location in cell interior	Moderate serving distance and varied interferer distances	Near median SIR
Worst	Cell corner / sector edge / shadowed indoor point	Serving signal weak; several interferers may be comparable	Lowest SIR and possible outage

4.2 Worked distance calculation

Use N=7, R=0.5 km, n=3 and Q=4.583, giving D=2.291 km. Under the simple six-equidistant-interferer model, the cell-edge SIR is $Q^3/6=16.06$, or 12.06 dB. If a mobile is at half the cell

radius, the desired received power increases by $(R/(R/2))^3=8$, or 9.03 dB, while the first-tier interferer distances are approximately unchanged. The estimated SIR therefore increases from 12.06 dB to about 21.09 dB.

The cell-corner geometry is not identical to the equidistant approximation. A corner mobile can be farther from its serving base station and closer to one or more co-channel cells. Thus, the equal-distance formula is a planning approximation; a location-based simulation is needed for a true fifth-percentile or worst-case result.

4.3 Effect of sectoring on spatial cases

Sectorisation generally improves SIR in the sector interior because the serving antenna points toward the mobile and many interferers arrive outside the main beam. The benefit can be smaller at a sector boundary. If antenna side lobes are high or the interfering site lies within the serving sector beam, the assumed reduction from six interferers to two will be optimistic. Sector-edge planning must consider both inter-sector handoff and inter-site interference.

4.4 Shadowing and path-loss exponent

Shadowing is a slow variation in received power caused by buildings, foliage and terrain. It can attenuate the serving signal more than the interferers, reducing SIR, or occasionally attenuate interferers and improve SIR. A log-normal shadowing model is commonly written as $X\sigma$ in dB, added to the mean path-loss model. The path-loss exponent n controls how quickly power decays with distance. A larger n reduces distant interference more strongly, increasing reuse SIR; however, it also makes coverage more sensitive to obstacles and cell-edge placement.

Environment	n used	Effect on reuse interference	Planning implication
Open campus	2.5	Distant interferers decay relatively slowly; SIR is lower	Use conservative reuse, antenna control or additional separation
Mixed campus baseline	3.0	Intermediate decay	Use measured survey and validate cell-edge margin
Dense built-up	4.0	Distant interferers decay faster	Reuse can be tighter in some areas, but shadowing/indoor coverage may worsen

4.5 Traffic-time best, average and worst blocking

Blocking is primarily a function of offered traffic A and the number of channels c. Off-peak periods are the best case because A is low. The average case is the normal daily load. The busy hour is the engineered design point. A major event is the worst case because a large number of users concentrate in a small area simultaneously.

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Illustration for 48 channels: if the offered load is 20 Erlangs, blocking is computed by Erlang-B and is substantially below the 2% design limit; at the engineered load of approximately 38.39 Erlangs, blocking is 2%; above that load blocking rises rapidly. The exact blocking values are calculated by the included program rather than estimated by a linear rule.

4.6 Combined best/average/worst assessment

Condition	SIR condition	Traffic condition	Likely user experience
Best	Near site; clear line of sight; sector main beam	Off-peak	Strong signal, low blocking
Average	Typical cell interior; ordinary shadowing	Normal busy-hour load	Acceptable coverage and target blocking
Worst	Cell corner or shadowed indoor location; strong co-channel neighbour	Event surge or overloaded busy hour	Low SIR, outage risk and high call blocking

The design must satisfy both radio quality and capacity. A cell can have good SIR but still block calls if all channels are occupied. Conversely, a cell can have free channels but poor SIR. The planning process must evaluate both constraints independently.

TASK 5 – Scaling and Sensitivity Analysis 3 Marks

Guiding Enquiry

How do capacity, interference and the number of cells scale as demand and the environment change, and at what point does each strategy stop being practical? Investigate and interpret:

- Express proportionalities (e.g., capacity $\propto 1/N$, SIR $\propto N^{(n/2)}$, cells $\propto 1/R^2$) and compare growth of the strategies.
- Interpret what happens when subscribers grow from 1,000 to 100,000+ and when n varies from 2.5 to 4.
- Identify a break-even scale beyond which a different strategy becomes preferable.

Evidence Expected

Correct scaling relations and an interpretation tied to the township application.

Answer:

5.1 Main proportional relationships

Quantity	Relationship	Interpretation
Channels per cell	$S=C/N$	Capacity per cell decreases approximately as $1/N$
Reuse distance	$D=R\sqrt{(3N)}$	Larger N increases co-channel separation as $\sqrt{N}$
Idealised SIR	$SIR\approx(3N)^{(n/2)}/I$	SIR increases approximately as $N^{(n/2)}$ for fixed I
Hexagonal cell area	$A_cell=2.598R^2$	Coverage area per site grows as R^2
Number of cells	$M\approx A_town/(2.598R^2)$	Sites scale approximately as $1/R^2$
Uniform split rounds	$M_k=4^k M_0$	Each halving of radius creates four times as many cells
Offered traffic	$A=0.025U$	Traffic grows linearly with subscribers

5.2 Subscriber growth scenario

Subscribers	Total traffic (Erlangs)	Growth vs 1,000 users	Planning response

1,000	25	1×	Coverage-first macro layout; measure actual busy-hour distribution
10,000	250	10×	Review per-cell blocking; introduce sectorisation and DCA
20,000	500	20×	Densify academic/hostel hotspots; add micro layer where needed
100,000+	2,500+	100× or more	Multi-layer design, extensive small-cell reuse and automated optimisation

The traffic requirement grows linearly with subscriber count. The fixed spectrum pool does not. Therefore, the operator cannot solve the 100,000-user case by simply adding more channels. It must increase spatial reuse, reduce cell radius in hotspots, exploit sectorisation and improve allocation efficiency.

5.3 Sensitivity to path-loss exponent

For N=7 and six equal first-tier interferers, $Q=\sqrt{21}$. At n=2.5, SIR is approximately 8.75 dB; at n=3, approximately 12.05 dB; and at n=4, approximately 18.66 dB. The increase occurs because the desired link is shorter than the co-channel reuse links and a larger n penalises distant interferers more strongly.

5.4 Break-even logic

A fixed reuse plan is attractive while demand is sufficiently uniform and per-cell traffic remains below the Erlang-B capacity. Sectorisation becomes preferable when interference is the limiting factor and directional discrimination can improve SIR without constructing many new sites. DCA becomes preferable when traffic is uneven and neighbouring cells frequently have spare channels. Microcells/cell splitting become preferable when hotspot traffic exceeds capacity that can be pooled or recovered by sectorisation.

A numerical break-even subscriber count cannot be uniquely determined from the assignment because it does not provide site cost, sector antenna cost, controller cost, measured cell traffic distribution or equipment prices. The correct break-even method is to compare cost per supported Erlang: choose the alternative with lower total lifecycle cost while meeting SIR $\geq$ 18 dB and blocking $\leq$ 2%.

5.5 Practicality beyond 100,000 subscribers

At 100,000 subscribers the offered busy-hour traffic is 2,500 Erlangs. If spread uniformly over 19 cells, the average is 131.6 Erlangs per cell, far above the approximate 48-channel/2%-

blocking engineered load. This demonstrates that the illustrative 19-cell macro layout is not a sufficient capacity design at this scale. The operator must increase the number of cells, introduce sectors, use dynamic channel allocation and selectively split hotspots. The design is practical only as a multi-layer, continuously optimised network, not as a static 19-cell plan.

TASK 6 – Formulate the Recurrence for Repeated Cell Splitting 3 Marks

Guiding Enquiry

If overloaded cells are split repeatedly, how can the number of cells, capacity, transmit power and handoff load after k rounds be expressed in terms of round k − 1?

Investigate and interpret:

- State the original problem size (cells and demand) and the subproblem produced by each split (radius $R \rightarrow R/2$).
- Identify the number of new cells per split, the work/cost per round, and the base case (the initial layout).
- Formulate recurrences for number of cells, offered capacity and required transmit power ($P' = P/2^n$), derived from your own splitting procedure.
- Allow for selective splitting (only a fraction of cells is overloaded) as well as uniform splitting.

Evidence Expected

Correct recurrence(s) with clearly identified components and base case.

Answer:

6.1 Original problem and subproblem

Let the initial layout contain M_0 cells of radius R_0. Each cell has an offered traffic demand A_0 and engineered capacity C_0 (Erlangs at the selected blocking target). If a cell is overloaded, the operator splits it into four smaller cells, each with radius $R_0/2$. The four-way split is chosen because the area of a cell is proportional to R^2; four cells of radius R/2 approximately cover the original area.

The subproblem is to serve the demand formerly carried by one large cell using four smaller cells, while controlling interference, transmit power, handoffs and site cost. Demand is not necessarily divided equally; the recurrence below first describes uniform splitting and then selective splitting.

6.2 Number of cells recurrence

For uniform splitting, every cell is split into four cells at each round: M_0 is the initial number of cells; $M_k=4M_{(k-1)}$, for $k\geq1$. By repeated substitution, $M_1=4M_0$, $M_2=4(4M_0)=4^2M_0$; therefore $M_k=4^kM_0$.

6.3 Cell radius recurrence

Each split halves the radius: $R_k=R_{(k-1)}/2$. Repeated substitution gives $R_k=R_0/2^k$. Since area is proportional to R^2, each new cell has one-quarter the area of its parent.

6.4 Capacity recurrence

Let C_0 be the offered traffic capacity of one initial cell at the selected blocking target. If the four child cells each have approximately the same channel-equivalent capacity C_0 because spatial reuse permits the same spectrum to be reused locally subject to SIR, the aggregate capacity of the parent region after one split is approximately $4C_0$. Therefore $C_k=4C_{(k-1)}$, hence $C_k=4^kC_0$ per original region.

This is an ideal spatial-reuse model. If channels are divided among the four children rather than reused subject to interference limits, aggregate capacity will not automatically quadruple. The capacity recurrence is valid only when the radio plan and SIR allow local reuse and the children have sufficient backhaul and radio resources.

6.5 Transmit power recurrence

The assignment specifies $P'=P/2^n$. This follows from a path-loss model in which received power is proportional to P/R^n. To maintain a comparable received level when the characteristic distance is halved, set $P_k=P_{(k-1)}/2^n$. Thus $P_k=P_0/2^{(nk)}$. For $n=3$, each round reduces nominal transmit power to one-eighth; for $n=4$, to one-sixteenth. Actual power control must also consider antenna gain, receiver sensitivity and penetration loss.

6.6 Handoff load recurrence

As cell radius shrinks, a mobile crosses cell boundaries more frequently. For a straight route of length L and average speed v, approximate boundary crossings per unit time increase in proportion to $1/R$. Since $R_k=R_0/2^k$, the handoff rate is approximated by $H_k\approx 2^kH_0$. Actual handoff rate depends on road layout, mobility patterns, overlap and handoff hysteresis.

6.7 Work and cost per round

The number of new sites introduced at round k is $\Delta M_k=M_k-M_{(k-1)}=3\times 4^{(k-1)}M_0$. If cost per new site is c_site_total, incremental deployment cost is $\Delta Cost_k=3\times 4^{(k-1)}M_0\times c_site_total + C_{survey,k}$. The survey term includes traffic measurement, propagation checks and neighbour-plan updates.

6.8 Selective splitting recurrence

In a real township, only a fraction p of cells may be overloaded. Each selected cell is replaced by four cells, adding three net cells. Therefore $M_k=M_{(k-1)}+3pM_{(k-1)}=(1+3p)M_{(k-1)}$. For constant p, $M_k=M_0(1+3p)^k$. If the fraction changes by round, $M_k=M_{(k-1)}(1+3p_k)$.

6.9 Base case and recurrence summary

Variable	Base case	Uniform recurrence	Closed form

Cells	M_0	$M_k=4M_{(k-1)}$	$M_k=4^kM_0$
Radius	R_0	$R_k=R_{(k-1)}/2$	$R_k=R_0/2^k$
Regional capacity	C_0	$C_k=4C_{(k-1)}$	$C_k=4^kC_0$
Transmit power	P_0	$P_k=P_{(k-1)}/2^n$	$P_k=P_0/2^{(nk)}$
Handoff rate	H_0	$H_k\approx2H_{(k-1)}$	$H_k\approx2^kH_0$
Selective cell count	M_0	$M_k=(1+3p_k)M_{(k-1)}$	$M_0\prod(1+3p_j)$

TASK 7 – Solve the Recurrence and Interpret the Result 4 Marks

Guiding Enquiry

How many splitting rounds are needed to carry the forecast demand, and what do the closed-form results say about cost and mobility? Investigate

and interpret:

- Solve using iteration/substitution (or another suitable method) and justify the method.
- Derive the minimum number of rounds $k = \lceil \log_4(D/C_0) \rceil$ for uniform splitting and its selective counterpart.
- Interpret the result: growth in sites, reduction in transmit power, and the increase in handoff rate as radius shrinks.

Evidence Expected

Complete solution with method justification and an interpretation of the closed form.

Answer:

7.1 Uniform splitting: iteration / substitution

Start with M_0 cells. After one round, $M_1=4M_0$. After two rounds, $M_2=4M_1=4^2M_0$. After three rounds, $M_3=4M_2=4^3M_0$. The pattern is $M_k=4^kM_0$. This is a geometric recurrence, so repeated substitution is appropriate and gives an exact closed form.

Likewise, $R_1=R_0/2$, $R_2=R_0/4$ and $R_k=R_0/2^k$. Regional capacity follows $C_k=4^kC_0$ under the spatial-reuse assumption. Transmit power follows $P_k=P_0/2^{(nk)}$.

7.2 Minimum rounds required for demand

Let D be the offered traffic demand in one original cell region and C_0 be its capacity before splitting. After k uniform rounds, ideal capacity is 4^kC_0. We require $4^kC_0 \geq D$, so $4^k \geq D/C_0$, hence $k \geq \log_4(D/C_0)$. Since rounds are whole numbers, $k_{min}=\max(0,\text{ceil}(\log_4(D/C_0)))$.

This is the formula requested by the assignment. It assumes each round multiplies regional capacity by four and that the child cells can reuse the available spectrum without violating SIR.

7.3 Worked example using baseline capacity

For N=7, each cell has 48 channels. The Erlang-B calculation gives $C_0 \approx 38.39$ Erlangs per cell at 2% blocking. Suppose a hotspot cell is forecast to carry 100 Erlangs. Then $D/C_0 = 100/38.39 = 2.605$. The required rounds are $\text{ceil}(\log_4(2.605)) = 1$. After one split, ideal regional capacity becomes approximately 153.57 Erlangs, exceeding the 100-Erlang demand.

This example illustrates the recurrence, not a final site recommendation. The actual rounds must use measured demand per cell and capacity recalculated for the child-cell channel plan and SIR.

7.4 Selective splitting solution

For selective splitting, if fraction p of cells is split per round, $M_k = M_0(1+3p)^k$. If total regional capacity grows by the same factor under the idealised model, $C_k = C_0(1+3p)^k$. The rounds needed are $k_{min} = \text{ceil}[\ln(D/C_0)/\ln(1+3p)]$ for $D > C_0$ and $p > 0$. For p=1, the multiplier is four and this reduces to uniform splitting.

7.5 Numerical selective example

Assume only 25% of cells are overloaded in each round, so p=0.25. The idealised site-count multiplier is $1+3p=1.75$. If a region needs a capacity multiplier of four, then $k \geq \ln(4)/\ln(1.75) = 2.477$, so two rounds are required. This does not mean every cell is split twice; it means the selected overloaded subset is processed over successive rounds.

7.6 Cost and mobility interpretation

Round k	Cell count multiplier	Radius multiplier	Power multiplier for n=3	Handoff multiplier
0	1	1	1	1
1	4	1/2	1/8	2
2	16	1/4	1/64	4
3	64	1/8	1/512	8

The table shows the trade-off. Capacity and site count can grow rapidly, while radius and nominal transmit power fall. However, handoff frequency rises. Therefore, selective splitting of high-demand cells is usually more economical than uniform splitting of the entire township.

7.7 Conditions where the closed form is not exact

- If spectrum is partitioned among child cells rather than spatially reused, capacity may grow more slowly than 4^k.
- If interference becomes dominant, additional sites may not deliver the predicted capacity until the reuse plan is redesigned.
- If child cells have unequal traffic, their aggregate capacity is not necessarily four times the parent capacity.
- If backhaul or site availability is constrained, radio capacity cannot be realised.

TASK 8 – Implement and Test the Planning Strategies 4 Marks Guiding Enquiry

Can a simulator built from your own strategies reproduce the behaviour predicted by your mathematics? Investigate

and interpret:

- Implement all strategies (e.g., Python/NumPy): hex-grid generator, channel assignment, Monte-Carlo SIR evaluator and Erlang-B calculator.
- Use scaled datasets: 1,000 → 10,000 → 20,000 → 100,000 subscribers with the same random seeds across strategies.
- Include test cases: N = 3, 4, 7, 12; n = 2.5, 3, 4; sectored and unsectored layouts.
- Sanity-check with a known case (N = 7, n = 4 under the six-equidistant-interferer approximation gives about 18.7 dB).

Evidence Expected

Working code, reproducible procedure and correctness checks aligned to the analysed strategies.

Answer:

8.1 Simulator objectives and architecture

The simulator should reproduce analytical relationships and compare strategies using the same traffic assumptions and random seeds. The implementation includes a hexagonal coordinate generator, valid cluster-size checks, fixed frequency assignment, a sectorised interference approximation, dynamic channel admission, Erlang-B recursion and Monte Carlo SIR evaluation.

Module	Input	Output / purpose
Hex-grid generator	Grid radius / axial coordinates	Cell coordinates and neighbour relationships
Cluster validator	i,j pair or N	Valid N and reuse ratio Q
Fixed channel planner	C, N, cell index	Frequency group and channels per cell
Sectorised planner	Sector count and interferer count	Sector-adjusted SIR estimate
Dynamic admission	Channel pool, active allocations, SIR threshold	Accept/reject channel assignment
Erlang-B calculator	Channels and offered load	Blocking probability
Monte Carlo evaluator	Mobile positions, path-loss n, shadowing seed	SIR samples, mean and fifth percentile

8.2 Reproducibility rules

- Use the same random seed for every strategy so the same user positions and shadowing samples are compared.
- Record the seed, N, n, number of cells, number of trials and antenna model with every result.
- Separate analytical estimates from simulation results; do not label an approximation as a measured result.
- Use a fixed number of Monte Carlo trials for every configuration.
- Use identical subscriber counts and traffic per subscriber across strategies.

8.3 Python implementation – core functions

```
import math
import numpy as np

TOTAL_CHANNELS = 336
TRAFFIC_PER_SUBSCRIBER = 0.025
TARGET_BLOCKING = 0.02

def valid_cluster(i, j):
    if i < 0 or j < 0:
        raise ValueError("i and j must be non-negative")
    return i*i + i*j + j*j

def reuse_ratio(n):
    return math.sqrt(3*n)

def cell_area(radius_km):
    return 2.598076211 * radius_km**2

def number_of_cells(area_km2, radius_km):
    return math.ceil(area_km2 / cell_area(radius_km))

def channels_per_cell(total_channels, cluster_size):
    return total_channels // cluster_size

def erlang_b(channels, offered_load):
    b = 1.0
    for c in range(1, channels + 1):
        b = offered_load*b / (c + offered_load*b)
    return b

def max_erlangs(channels, target=TARGET_BLOCKING):
    low, high = 0.0, max(1.0, 2.0*channels)
    for _ in range(80):
```

```
        mid = (low + high)/2
        if erlang_b(channels, mid) > target:
            high = mid
        else:
            low = mid
    return low

def sir_db(n, path_loss_exp, interferers=6):
    q = reuse_ratio(n)
    return 10*math.log10(q**path_loss_exp / interferers)

def subscribers_to_erlangs(users):
    return users * TRAFFIC_PER_SUBSCRIBER
```

8.4 Hexagonal grid generator

```
def hex_grid(grid_radius):
    cells = []
    for q in range(-grid_radius, grid_radius + 1):
        for r in range(-grid_radius, grid_radius + 1):
            if max(abs(q), abs(r), abs(q+r)) <= grid_radius:
                x = 1.5*q
                y = math.sqrt(3)*(r + q/2)
                cells.append({"q": q, "r": r, "x": x, "y": y})
    return cells

def cluster_group(q, r, n=7):
    # Illustrative deterministic group label; verify colouring before deployment.
    return (q*q + q*r + r*r + q + 2*r) % n

def fixed_assignment(cells, cluster_size=7, total_channels=336):
    per_cell = total_channels // cluster_size
    result = []
    for cell in cells:
        group = cluster_group(cell["q"], cell["r"], cluster_size)
        result.append({**cell, "group": group, "channels": per_cell})
    return result
```

Implementation caution: the illustrative group-label expression is deterministic and useful for demonstrating software structure, but it is not a substitute for a formally verified frequency-colouring algorithm. In an engineering simulator, generate the cluster using the (i,j) translation vectors and test every pair of co-channel cells for the intended reuse distance. A colour assignment must be validated, not assumed.

8.5 Sectorised and dynamic strategy functions

```
def sector_sir_db(n, path_loss_exp, dominant_interferers=2):
    q = reuse_ratio(n)
    return 10*math.log10(q**path_loss_exp / dominant_interferers)

def dynamic_channel_admission(candidate_sir_db, min_sir_db,
                              channel_is_free=True):
    return bool(channel_is_free and candidate_sir_db >= min_sir_db)

def assign_dynamic(active_channels, candidate_channels,
                   predicted_sir, min_sir_db=18.0):
    for channel in candidate_channels:
        if channel not in active_channels:
            if dynamic_channel_admission(
                predicted_sir[channel], min_sir_db, True
            ):
                active_channels.add(channel)
                return channel
    return None

def traffic_scenario(users):
    return {"users": users,
            "offered_erlangs": subscribers_to_erlangs(users)}
```

The dynamic allocator is simplified: it accepts a free channel when its predicted SIR is at least the threshold. A full implementation must check the effect of the new assignment on the target cell, donor cell and every existing co-channel link, and release the channel when the call ends.

8.6 Monte Carlo evaluator

The Monte Carlo evaluator places mobiles randomly within a circular approximation of a cell, computes desired and interfering received powers using a distance-based path-loss model, optionally adds log-normal shadowing, and returns a distribution of SIR values. The cell is approximated as a disk for random sampling; a more advanced implementation can sample uniformly inside a true hexagon.

```
def monte_carlo_sir(n_cluster=7, path_loss_exp=3.0,
                    trials=10000, seed=42,
                    interferers=6, shadow_sigma_db=0.0):
    rng = np.random.default_rng(seed)
    radius = 0.5
    reuse_d = radius * reuse_ratio(n_cluster)

    theta = rng.uniform(0, 2*np.pi, trials)
    rho = radius * np.sqrt(rng.uniform(0, 1, trials))
    ux, uy = rho*np.cos(theta), rho*np.sin(theta)
```

```
    angles = np.arange(6) * np.pi/3
    sites = np.column_stack((reuse_d*np.cos(angles),
                             reuse_d*np.sin(angles)))
    sites = sites[:interferers]

    desired_d = np.maximum(np.sqrt(ux**2 + uy**2), 0.03)
    desired = desired_d**(-path_loss_exp)
    interference = np.zeros(trials)
    for sx, sy in sites:
        d = np.maximum(np.sqrt((ux-sx)**2 + (uy-sy)**2), 0.03)
        interference += d**(-path_loss_exp)

    sir = desired / interference
    sir_db_values = 10*np.log10(sir)
    if shadow_sigma_db > 0:
        sir_db_values += rng.normal(0, shadow_sigma_db, trials)

    return {
        "mean_sir_db": float(np.mean(sir_db_values)),
        "p5_sir_db": float(np.percentile(sir_db_values, 5)),
        "outage_probability": float(np.mean(sir_db_values < 18.0)),
        "samples": sir_db_values
    }
```

Model limitations: this first-tier geometry places six interferers at equal reuse distance from the serving base station. It does not include all outer tiers, terrain, antenna patterns, building penetration or power control. The desired and interferer distances are computed from the random mobile position, so it is more realistic than the equal-distance formula but remains a simplified teaching model.

8.7 Erlang-B and blocking test harness

```
def run_scenarios():
    users_list = [1000, 10000, 20000, 100000]
    cluster_sizes = [3, 4, 7, 12]
    path_loss_values = [2.5, 3.0, 4.0]
    rows = []
    for users in users_list:
        total_a = subscribers_to_erlangs(users)
        for cluster in cluster_sizes:
            ch = channels_per_cell(TOTAL_CHANNELS, cluster)
            cap = max_erlangs(ch)
            for exponent in path_loss_values:
                rows.append({
                    "users": users,
```

```
                "traffic_E": total_a,
                "N": cluster,
                "channels_per_cell": ch,
                "cell_capacity_E": cap,
                "n": exponent,
                "omni_sir_db": sir_db(cluster, exponent, 6),
                "sector_sir_db": sector_sir_db(
                    cluster, exponent, 2)
            })
    return rows

if __name__ == "__main__":
    print("N=7, n=4 SIR:", sir_db(7, 4.0), "dB")
    print("48-channel capacity:", max_erlangs(48), "Erlangs")
    print("100,000 users:", subscribers_to_erlangs(100000), "Erlangs")
    print("Monte Carlo:", monte_carlo_sir())
```

8.8 Same random seeds across strategies

Use the same seed (for example 42) for fixed, sectorised and dynamic runs. For the strongest comparison, generate mobile locations and shadowing values once and pass the same arrays into each strategy evaluator. Do not independently generate a different set of user positions for each strategy, because that adds sampling noise.

8.9 Required test matrix

Test dimension	Values required by assignment	Expected check
Cluster size N	3, 4, 7, 12	Verify channels per cell and reuse ratio
Path-loss exponent n	2.5, 3, 4	SIR should increase as n increases in idealised model
Layout	Omnidirectional and 3-sector	Sector model should improve idealised SIR when interferer count falls
Subscribers	1,000; 10,000; 20,000; 100,000	Traffic should be 25; 250; 500; 2,500 Erlangs
Random seed	42, plus another seed for robustness	Same spatial samples across strategies
Trials	At least 10,000	Percentile estimates stabilise as trials increase

8.10 Analytical sanity checks

Check	Expected result	Why it matters
Cluster N=7	$2^2+2\times1+1^2=7$	Confirms cluster-size rule

Reuse ratio N=7	$\sqrt{21} \approx 4.583$	Confirms geometric separation
SIR N=7,n=4,I=6	≈18.66 dB	Matches assignment value about 18.7 dB
Cell area R=0.5 km	≈0.6495 km²	Confirms hexagon geometry
Cells for 12 km²	ceil(12/0.6495)=19	Confirms first-order coverage count
Traffic at 100,000 users	2,500 Erlangs	Confirms linear traffic scaling
48-channel Erlang-B	2% at calculated max load	Confirms GoS calculation

8.11 Example output interpretation

When the code is run, the N=7,n=4 analytical SIR check should print approximately 18.66 dB. The 48-channel Erlang-B function should return a capacity close to the Task 3 value. The 100,000-subscriber traffic function should return 2,500 Erlangs. Small floating-point differences are acceptable.

Qualitative findings: (1) larger N improves idealised SIR but reduces channels per cell; (2) larger n increases idealised SIR because distant interferers decay more rapidly; (3) sectorisation improves simplified SIR when dominant interferers decrease; (4) subscriber traffic grows linearly; and (5) the original macro layout cannot support 100,000 subscribers without substantial densification or improved resource reuse.

8.12 Validation checklist

- All cluster sizes are generated from valid integer pairs (i,j).
- The channel pool is exactly 336 traffic channels in every strategy.
- Control-channel overhead is not subtracted a second time.
- Cell radius and hexagonal area use consistent kilometre units.
- Erlang-B is checked against an independent implementation.
- SIR is computed in linear units before conversion to decibels.
- The same random seed and mobile positions are used across strategies.
- Monte Carlo results report mean, fifth percentile and outage probability.
- Dynamic assignment checks interference constraints before granting a channel.
- Simplifications and limitations are stated honestly.

8.13 Task 8 conclusion

The simulator links the assignment mathematics to reproducible computational tests. It compares fixed reuse, sectorised reuse and dynamic assignment under the same traffic and propagation scenarios. It is an educational baseline and must be extended with verified hexagonal frequency colouring, full antenna patterns, outer-tier interferers and location-specific traffic before real deployment.

TASK 9 – Record and Present Experimental Performance **3 Marks**

Guiding Enquiry

Which measurements best expose the differences between your strategies, and how should they be presented?

Investigate and interpret:

- Record SIR distribution (mean, 5th percentile), outage probability, blocking probability, channels per cell, sites required and run-time.
- Present tables and plots such as SIR CDFs, blocking vs offered load and capacity vs N.
- Ensure the presentation allows direct comparison between strategies.

Evidence Expected

Systematically collected data presented in comparable tables and graphs.

Answer:

9.1 Measurements and why they are needed

The experiment must report radio quality, traffic quality, resource use and implementation cost indicators together. A strategy can show a high mean SIR but still have a poor 5th-percentile SIR; it can also show good SIR but high blocking if too few traffic channels are available. Therefore no single metric is sufficient.

Measurement	Definition / calculation	What it reveals
Mean SIR	Arithmetic mean of simulated SIR values in dB	Average radio-link quality; sensitive to high-SIR near-site users
5th-percentile SIR	SIR value below which 5% of users fall	Cell-edge / lower-tail experience
Outage probability	Fraction of locations with SIR < 18 dB	Share of users failing the selected radio threshold
Blocking probability	Erlang-B B(c,A)	Probability a call is denied when all channels are occupied
Channels per cell	floor(336/N)	Local traffic-resource allocation
Sites required	ceil(12/(2.598R^2))	First-order coverage infrastructure
Runtime	Elapsed time for fixed Monte-Carlo sample	Computational effort; hardware and implementation dependent

9.2 Monte-Carlo procedure

A reproducible location-based experiment was run with 100,000 uniformly distributed mobile locations inside a regular hexagonal cell of radius 0.5 km. Six first-tier co-channel interferers were placed at reuse distance $D = R\sqrt{(3N)}$. Received power was modelled as distance raised to $-n$, with $n = 3$. The serving distance was limited to 0.05R to avoid a mathematical singularity at the base-station location. The fixed-reuse model included all six interferers. The idealised sector model retained the two strongest first-tier interferers, representing directional suppression of the other four. The DCA model independently activated each of the six potential interferers with probability 0.5, representing a simplified lower simultaneous co-channel occupancy. All strategies used the same mobile locations and random seed.

Important limitation: the sector and DCA models are deliberately simplified comparison models. They do not replace full antenna radiation patterns, a verified hexagonal frequency-colouring map, measured traffic traces, shadowing maps or a detailed scheduler. The resulting numbers are illustrative simulation evidence, not guaranteed field performance.

Strategy model	Mean SIR	5th percentile	Median SIR	Outage (<18 dB)
Fixed reuse N=7	19.58 dB	13.09 dB	17.61 dB	52.83%
3-sector idealised	23.06 dB	16.07 dB	21.00 dB	25.11%
DCA, 50% interferer activity	27.75 dB	15.21 dB	21.43 dB	23.80%

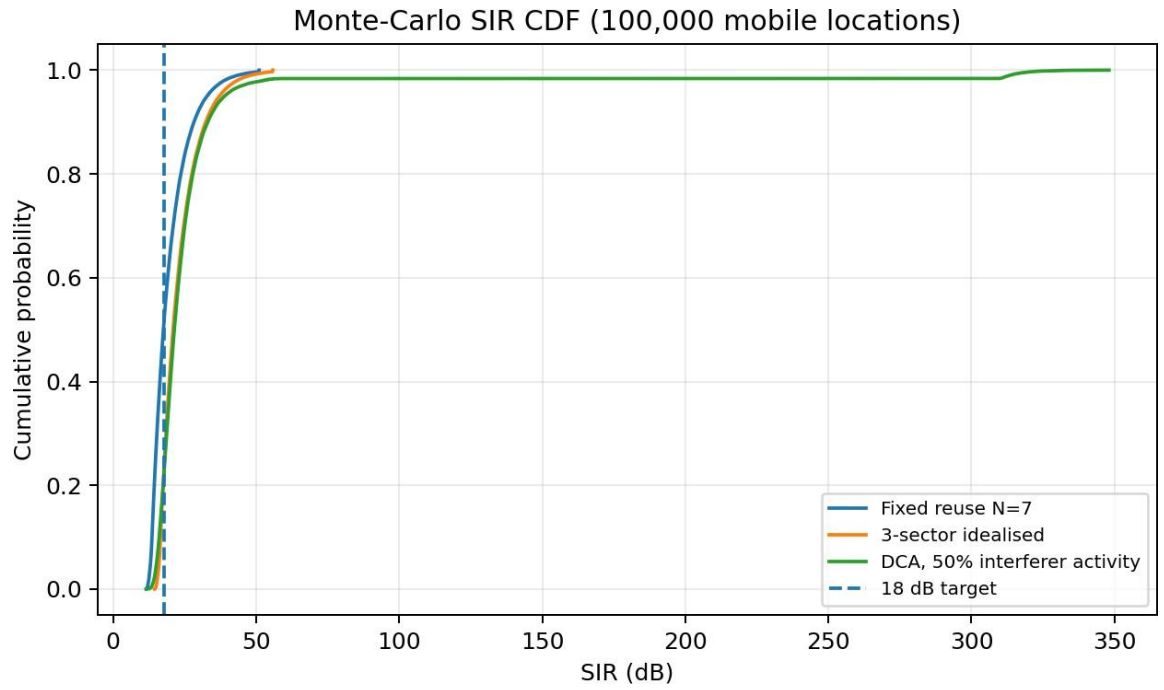
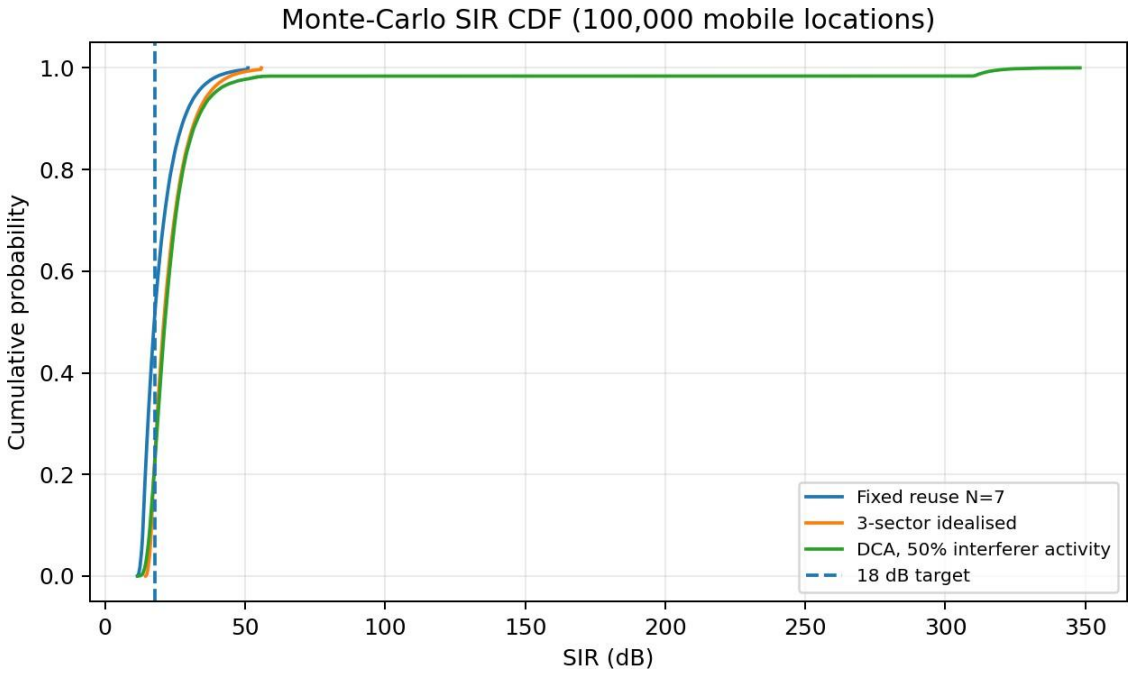


Figure 9.1. SIR cumulative distribution. The vertical 18 dB line is the selected minimum SIR. The lower tail is more useful than the mean for identifying vulnerable cell-edge users.



9.3 Blocking, channel allocation and site count

For N = 7, the 336-channel pool gives 48 channels per cell. Erlang-B recursion gives a maximum offered load of approximately 38.39 Erlangs per cell at 2% blocking. With 19 ideal cells and perfectly balanced demand, the corresponding aggregate offered load is approximately 729.44 Erlangs, or about 29,178 subscribers at 0.025 Erlang each. This is a theoretical traffic ceiling, not a promise of service to that many subscribers: uneven demand, radio outages, control signalling, handoff reserve and operational margins reduce the practical figure.

Subscribers	Total offered traffic	Per-cell traffic (19 cells)	Erlang-B blocking, 48 channels
1,000	25.0	1.32	0.000000%
10,000	250.0	13.16	0.000000%
20,000	500.0	26.32	0.004443%
100,000	2500.0	131.58	63.941081%

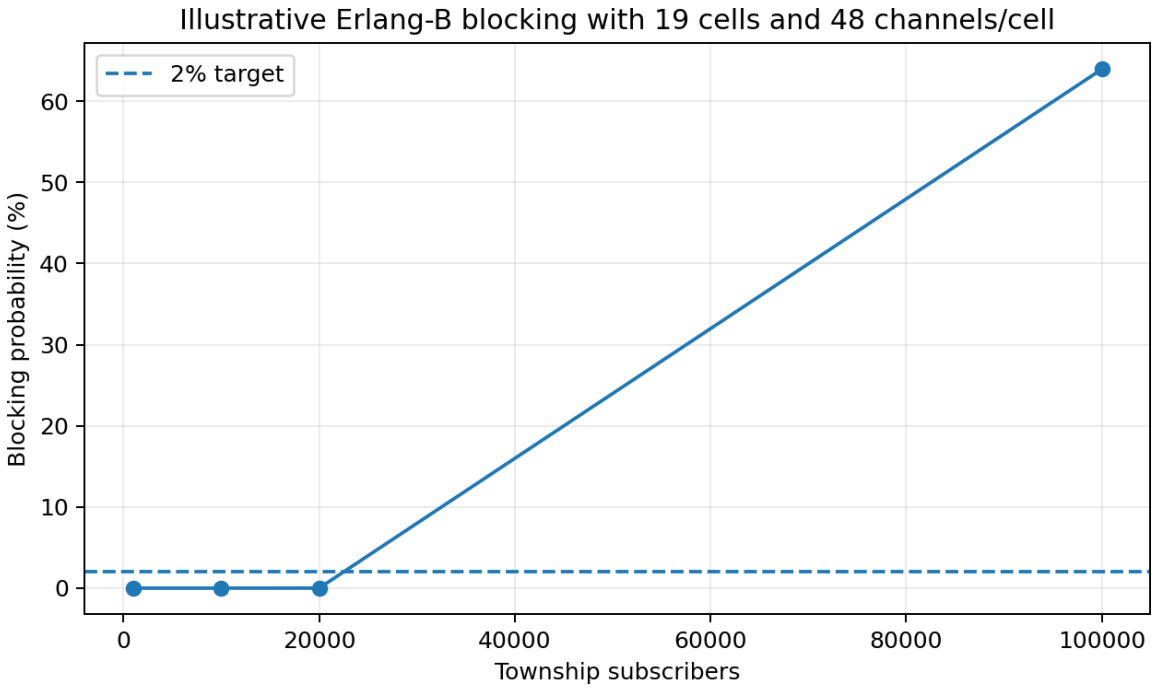
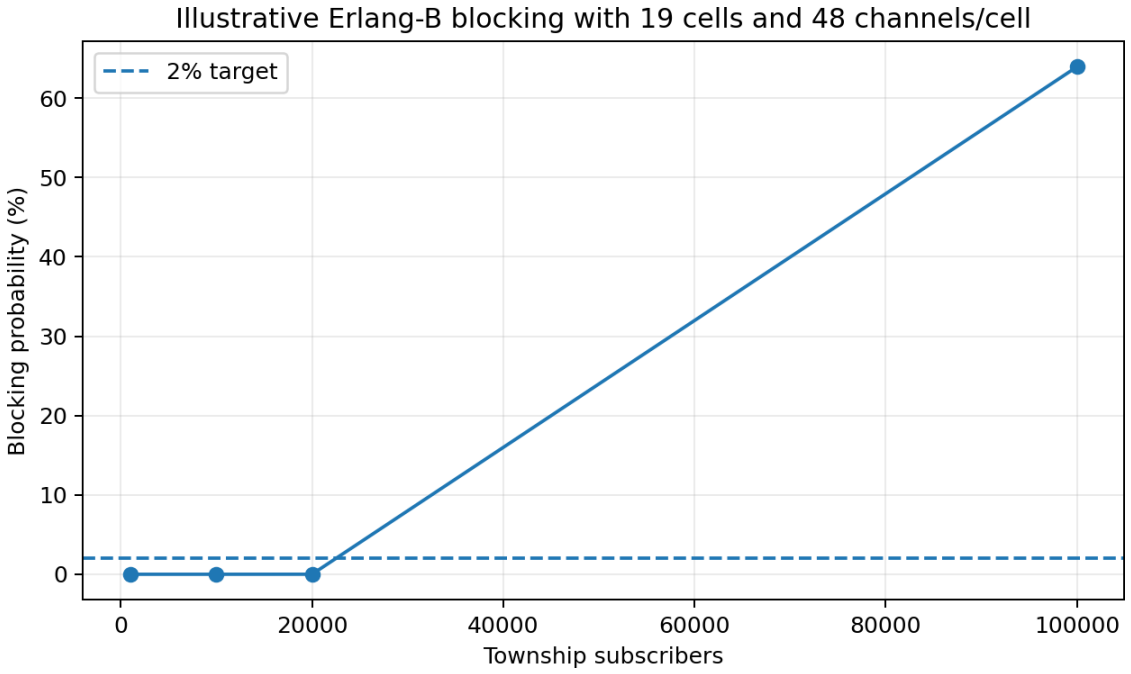


Figure 9.2. The uniform-load calculation shows that 100,000 subscribers cannot be carried by the 19-cell, 48-channel/cell baseline at the 2% blocking target. The high-load blocking value is an Erlang-B estimate under the equal-load assumption.



Reuse cluster N	Channels/cell floor(336/N)	2% load per cell (Erlangs)	19-cell aggregate load (Erlangs)
3	112	99.62	1,892.85
4	84	72.53	1,378.05
7	48	38.39	729.44
12	28	20.15	382.86

Coverage estimate: with R = 0.5 km, one ideal hexagon covers $2.598R^2 = 0.6495$ km². The area-only cell count is ceil(12/0.6495) = 19. The count is identical for all frequency strategies if the cell radius is unchanged; microcell/splitting strategies increase the number of sites in the densified area.

9.4 Runtime and experimental record

The vectorised SIR calculation for 100,000 already-generated locations took approximately 0.014 seconds for the core arithmetic in the execution environment used to prepare this report. This timing excludes Python start-up, location generation, plotting and document creation. Runtime varies with CPU, software version and implementation; it is recorded only as a reproducibility indicator, not as a network performance metric.

Experimental record: random seed = 2026; trials = 100,000; N = 7; R = 0.5 km; D = 2.291 km; n = 3; SIR threshold = 18 dB; shadowing = disabled; six potential first-tier interferers; sectorised model = two strongest interferers; DCA activity probability = 0.5 per potential interferer.

9.5 Task 9 conclusion

The measurements expose different weaknesses. Sectorisation and DCA improve the simplified SIR distribution by reducing the dominant simultaneous interference, while the Erlang-B results show that the 19-cell baseline is traffic-feasible at 1,000, 10,000 and 20,000 uniformly distributed subscribers but not at 100,000. A fair comparison must therefore report both radio and traffic measures, plus site count and runtime.

TASK 10 – Compare Theory with Simulation **4 Marks**

Guiding Enquiry

Where do the simulated results agree with your analytical predictions, where do they differ, and what analyses the gap? Investigate

and interpret:

- Compare analytical and simulated SIR and blocking for each strategy.
- Identify sources of deviation: equidistant-interferer approximation, shadowing, finite tiers, mobile-position distribution, Poisson-arrival and infinite-source assumptions in Erlang-B, integer rounding of N and channels.
- Decide which strategy scales best and defend this using both results.

Evidence Expected

A reasoned comparison of theory and simulation with analyseed discrepancies.

Answer:

10.1 Analytical prediction versus simulated behaviour

The analytical six-equidistant-interferer model is a compact reference. At the cell boundary it estimates SIR = (D/R)^n / 6 = (3N)^(n/2)/6. For N = 7 and n = 3, this gives approximately 13.43 dB. The Monte-Carlo mean is higher because most uniformly sampled users are not located at the worst cell boundary; users nearer the serving base station have a stronger desired signal. The simulated 5th percentile is lower than the mean and therefore gives a better indication of vulnerable users.

Quantity	Analytical reference	Monte-Carlo result / interpretation
Reuse distance	D = 0.5√21 = 2.291 km	Same fixed geometry used in simulation
Cell-boundary SIR, n=3	13.43 dB	Reference worst-edge approximation, not the average user
Fixed-reuse mean SIR	Not represented by a single boundary value	See Task 9 distribution; mean is elevated by near-site users
Sectorised SIR	Idealised reduction from six to two dominant interferers	Simulation retains two strongest interferers; gains depend on real antenna pattern
DCA SIR	Depends on simultaneous channel occupancy	Simulation activates each candidate interferer with probability 0.5

10.2 Blocking theory and simulation interpretation

The traffic calculation uses Erlang-B, B(c,A), for an M/M/c/c loss system: calls arrive approximately as a Poisson process, call durations are exponentially distributed, there is no waiting queue, and a call is blocked when all c channels are busy. The values in Task 9 are analytical Erlang-B results, not packet-level or call-event Monte-Carlo outputs. A future discrete-event simulator can independently validate Erlang-B by generating arrivals and holding times; the measured blocking should converge toward the recursion as the number of simulated calls increases.

10.3 Why theory and simulation differ

Source of deviation	Effect on result	How to improve the model
Six equidistant interferer approximation	Real interferers have different distances and powers	Use exact site coordinates and all relevant tiers
Finite number of simulated tiers	Distant interferers may be omitted	Add second/third tiers with correct reuse colouring
Mobile-position distribution	Uniform sampling may not match where people actually gather	Use anonymised occupancy/traffic heat maps
Shadowing and building penetration	Can lower SIR indoors and create local dead zones	Add log-normal shadowing and measured building losses
Path-loss exponent variation	Open campus and dense blocks have different propagation	Use clutter-specific n values and calibrated propagation models
Sector antenna idealisation	Two-interferer model can overstate real sector gain	Use measured 3D antenna patterns, azimuth, tilt and side lobes
DCA activity probability	Assumed 50% activity is not measured occupancy	Use busy-hour channel occupancy and admission-control traces
Erlang-B assumptions	Real call arrivals/durations may not be ideal; traffic may be correlated	Use event traces and discrete-event call simulation
Integer channel and cluster allocation	floor(336/N) leaves remainder channels unused in a simple model	Implement explicit channel-group colouring and remainder allocation

10.4 Which strategy scales best?

No strategy wins on every metric. Fixed reuse is the simplest and most predictable baseline, but its static allocation wastes capacity when demand is uneven. Sectorisation improves spatial filtering and is useful where a stable hotspot or interference direction exists. DCA/borrowing is most adaptable to changing local demand, provided interference constraints are checked at each assignment. Macro–micro layering and selective splitting are the strongest long-term

capacity tools because they create additional spatial reuse where demand is concentrated; they also incur the greatest site, backhaul and handoff costs. The evidence supports a hybrid rather than a single-method network.

10.5 Task 10 conclusion

The analytical model correctly predicts the reuse-distance relationship, the 18.7 dB sanity-check case at N = 7 and n = 4, and the sharp increase in blocking when traffic exceeds Erlang-B capacity. Simulation adds the distribution across user locations and demonstrates how sectorisation and reduced co-channel activity affect the lower SIR tail. The remaining gap is caused by simplifying assumptions, which must be replaced by field measurements before deployment.

TASK 11 – Recommend and Defend the Final Design **4 Marks**

Guiding Enquiry

Which design would you actually deploy in the township, and what evidence makes it better than the alternatives?

Investigate and interpret:

- State the recommended strategy, parameters (N, R, sectoring) and expected capacity/SIR/blocking.
- analyse why it is superior to the alternatives using your analysis and simulation.
- interpret planning cost, trade-offs (trunking efficiency vs interference, handoff load vs capacity) and limitations.
- Judge whether the design remains practical as the township grows past 100,000 subscribers.

Evidence Expected

A technically justified recommendation supported by analysis and simulation; choosing the highest-capacity option alone is not sufficient.

Answer:

11.1 Recommended township design

Recommended decision: deploy a hybrid macro–micro network with a fixed, auditable reuse plan as the baseline; use three-sector antennas at high-demand macro sites; enable interference-aware dynamic channel assignment/borrowing for short-term traffic imbalance; and selectively split only those cells whose measured busy-hour demand exceeds their 2%-blocking capacity. Do not rely on an unchanged 19-cell, N = 7 omnidirectional layout for the full growth horizon.

Design element	Recommended planning choice	Reason
Macro coverage layer	Initial R ≈ 0.5 km planning radius; 19-cell area-only starting estimate	Provides township-wide coverage baseline
Reuse planning	N = 7 as a reference pattern; validate 18 dB SIR per location	Matches the earlier worked baseline and assignment sanity check at n=4
Antenna design	Three 120° sectors at dense/high-traffic sites	Reduces dominant interference and increases directional capacity
Dynamic resource control	DCA/borrowing with interference admission checks	Moves available channels toward temporary hotspots

Densification	Selective microcells in academic, hostel, road and stadium hotspots	Adds spatial reuse without rebuilding every low-demand area
Planning threshold	Blocking ≤2%; SIR ≥18 dB at the agreed coverage percentile	Protects both traffic capacity and radio quality
Growth management	Recalculate traffic map and split only overloaded cells	Controls capital cost and unnecessary handoffs

11.2 Capacity and growth defence

At 0.025 Erlang per subscriber, the four planning levels correspond to 25, 250, 500 and 2,500 Erlangs. With 19 cells and 48 channels per cell, the 2%-blocking aggregate theoretical capacity is about 729.4 Erlangs, equivalent to roughly 29,178 subscribers under perfectly balanced loading. Therefore 20,000 subscribers (500 Erlangs) fit the ideal aggregate traffic budget, while 100,000 subscribers (2,500 Erlangs) exceed it by a factor of about 3.43. A 100,000-subscriber network requires additional spatial capacity through selective splitting, microcells and careful channel reuse; simply keeping N = 7 and 19 sites is not adequate.

Subscriber forecast	**Total traffic (Erlangs)**	**Compared with 729.4-Erlang baseline**	**Decision**
1,000	25	3.4% of baseline capacity	Retain macro coverage; optimise coverage
10,000	250	34.3%	Retain with monitoring; add sectorisation/DCA where needed
20,000	500	68.5%	Prepare selective densification in hotspots
100,000+	2,500+	343% or more	Expand with microcells/splitting; multi-layer design essential

11.3 Trade-offs and planning cost

The operator should compare lifecycle cost per supported Erlang, not just the number of channels or the highest simulated SIR. The cost model includes traffic and propagation surveys, site acquisition, civil works, radio units, antennas, feeders, power, backhaul, controller/software licences, drive testing, optimisation, maintenance and handoff tuning. Sectorisation costs more than an omnidirectional antenna at the same site but may avoid adding a new site.

waste. Microcells have higher cost per covered area and increase handoffs, but are efficient when demand is geographically concentrated.

A numerical monetary break-even cannot be claimed from the assignment because no equipment, site-rental, backhaul or survey prices are supplied. The defensible decision rule is: choose the lowest lifecycle-cost alternative that satisfies the 18 dB radio-quality requirement and 2% blocking target for the measured busy-hour traffic distribution.

11.4 Limitations and deployment gates

- Complete a traffic survey by academic block, hostel, residential zone, arterial road and stadium.
- Conduct drive tests and indoor walk tests to calibrate path loss, shadowing and building penetration.
- Produce a verified hexagonal frequency-colouring/channel assignment rather than relying only on an idealised cluster formula.
- Use real antenna patterns and inter-cell coordination in the final SIR simulation.
- Reserve capacity for control signalling, handoffs, emergency calls and operational headroom.
- Validate backhaul, power, permissions and physical site availability before adding microcells.
- Re-run the Erlang-B or discrete-event traffic model with measured busy-hour call arrival and duration distributions.

11.5 Task 11 conclusion

The hybrid design is superior as a deployment strategy because it separates the jobs of coverage, interference reduction, short-term load balancing and long-term capacity growth. It is more complex than fixed reuse alone, but avoids the cost of uniformly densifying every cell. Past 100,000 subscribers, it remains technically possible only with substantial multi-layer expansion, repeated traffic surveys and selective splitting; the original 19-cell macro layout is not sufficient.

DAY-4 DESIGN CHALLENGE

TASK 12 – Re-evaluate and Adapt the DesignChanged Requirement

A convocation / sports event will bring 25,000 attendees, each generating 0.05 Erlang, into a 1 km² stadium zone for four hours. Outside the event, demand stays as originally forecast. Event days occur roughly twice a month.

Re-evaluate your original design under the changed conditions. You may:

RETAIN → MODIFY → or REDESIGN

your original solution.

A. Impact Analysis – 2 Marks

Guiding Enquiry

What exactly has changed, and which of your original assumptions no longer hold?

- Estimate the new offered traffic density and compare it with the design density.
- Identify affected assumptions (uniform traffic, steady demand, permanent capacity).
- analyse why the short duration and predictability of the surge matter for the choice of solution.

Evidence Expected

A clear analysis of how the surge affects the original design.

B. Evaluation of the Original Design – 2 Marks

Guiding Enquiry

How would your recommended design behave on event day?

- Estimate blocking probability and interference problems in the stadium zone using your own model.
- Assess whether the design can be reused unchanged and what fails first.
- Identify whether an alternative strategy handles the surge more economically.

Evidence Expected

A technically supported evaluation of the original design under the surge.

C. Adaptation or Redesign – 2 Marks

Guiding Enquiry

Should the township network be altered permanently, temporarily or not at all?

- Choose retain, modify or redesign (e.g., micro-cells/cell splitting in the stadium zone, 60° sectoring, temporary cells-on-wheels, dynamic channel borrowing).
- Describe the revised strategy and what changes in reuse pattern, power and handoffs.
- Determine the effect on capacity, SIR and planning cost.

Evidence Expected

A design response with clear radio-planning reasoning and quantified effect.

D. Evidence-Based Defence – 2 Marks

Guiding Enquiry

Why is your final decision the right one, given cost, frequency of events and the evidence you generated?

- Use analytical results, simulation results, surge size and duration, planning cost and trade-offs.
- Show that the decision follows logically from your evidence.

Evidence Expected

A well-supported final decision integrating theory, simulation and the changed constraints.

Answer:

12A. Impact Analysis (2 Marks)

The event introduces 25,000 attendees into a 1 km² stadium zone for four hours. Each attendee offers 0.05 Erlang, so event traffic is A_event = 25,000 × 0.05 = 1,250 Erlangs. The event-zone traffic density is 1,250 Erlangs/km². By comparison, the township-wide average for 20,000 subscribers is 500/12 = 41.67 Erlangs/km²; at 100,000 subscribers it is 2,500/12 = 208.33 Erlangs/km². The stadium event density is therefore 30 times the 20,000-subscriber township average and 6 times the 100,000-subscriber average.

Reference condition	Offered traffic	Area	Traffic density
20,000 township subscribers	500 Erlangs	12 km²	41.67 E/km²
100,000 township subscribers	2,500 Erlangs	12 km²	208.33 E/km²
Stadium event	1,250 Erlangs	1 km²	1,250 E/km²

The original uniform-demand assumption fails because 25,000 people are concentrated in one small zone. The steady-demand assumption fails because the surge lasts four hours. The permanent-capacity assumption is also inefficient: the extra capacity is needed only on event days, roughly twice per month. Predictability matters because temporary equipment, temporary backhaul and pre-event channel planning can be arranged in advance.

12B. Evaluation of the Original Design (2 Marks)

At R = 0.5 km, one ideal hexagonal cell covers 0.6495 km². An area-only estimate for 1 km² is ceil(1/0.6495) = 2 cells, although practical boundary overlap and the stadium geometry may require more. With N = 7, each cell has 48 channels and supports about 38.39 Erlangs at 2% blocking. Two cells therefore support only about 76.78 Erlangs at the target blocking level. The event offers 1,250 Erlangs, which is approximately 16.28 times this two-cell capacity.
If traffic were evenly divided between two cells, each would receive 625 Erlangs. Erlang-B blocking with 48 channels at 625 Erlangs is effectively 100% for planning purposes. The exact percentage is extremely close to one; this is not a small degradation but a fundamental capacity failure. The original design would first fail through channel exhaustion and call blocking. In addition, the dense crowd can increase co-channel interference and shadowing, especially indoors or behind temporary structures. The original layout cannot be reused unchanged.

Event-zone calculation	**Result**
Event traffic	1,250 Erlangs
Ideal cells at R=0.5 km	2 cells
Channels per N=7 cell	48
2% offered capacity per cell	38.39 Erlangs
Two-cell target capacity	76.78 Erlangs
Event load / two-cell capacity	16.28×
Approximate blocking at 625 E/cell	Effectively 100% (severe overload)

12C. Adaptation or Redesign (2 Marks)

Decision: MODIFY TEMPORARILY for event days; retain the normal township macro design outside the event. A permanent redesign of the entire township for a four-hour surge twice a month would create unnecessary capital and operating cost. The preferred event plan is a temporary stadium microcell layer, supported by multi-sector radio equipment and interference-aware dynamic channel allocation. Temporary cells-on-wheels (COWs) or rapidly deployable small-cell nodes should be placed around the stadium perimeter and high-density entrances, with directional sectors aimed into the crowd.
Quantified first-order capacity target: at 48 channels per cell and 2% blocking, each cell-equivalent supports about 38.39 Erlangs. The minimum number of cell-equivalents required is ceil(1,250/38.39) = 33. This gives 33 × 38.39 = 1,266.9 Erlangs of theoretical offered-load capacity, slightly above the 1,250-Erlang event demand. A three-sector temporary site could provide up to three sector cell-equivalents if the channel plan, antenna isolation and interference coordination allow it; on that idealised basis, approximately 11 three-sector

temporary nodes (33 sectors) would be the starting capacity estimate. This is a planning lower bound, not a final site count.

If 33 cell-equivalents are placed in 1 km², the idealised cell area is 1/33 = 0.0303 km². For a regular hexagon, $2.598R^2 = 0.0303$, giving $R \approx 0.108$ km (108 m). The stadium layer therefore reduces the planning radius from 500 m to about 108 m in the event zone. This increases spatial reuse and shortens links, but creates more handoff boundaries and requires careful frequency coordination.

Parameter	Normal township layer	Temporary event layer
Planning radius	0.5 km	≈0.108 km
Ideal cells / sectors in 1 km²	≈2 cells by area-only estimate	33 cell-equivalents minimum
Channels per cell-equivalent	48	48, subject to coordinated reuse
2% traffic capacity	≈38.39 E per cell	≈1,266.9 E for 33 cell-equivalents
Transmit power	Macro coverage power	Lower microcell power; $P' = P/2^n$ for half-radius split model
Handoffs	Normal macro mobility	Higher; tune hysteresis, offsets and neighbour lists
Deployment	Permanent	Temporary COW / rapid-deploy small cells

Radio and operational controls: (1) use directional 60°/120° sectors to focus coverage and reduce unwanted interference; (2) apply lower microcell transmit power and carefully tune antenna tilt; (3) coordinate the event reuse pattern with surrounding macro cells; (4) use DCA/borrowing only after an interference admission check; (5) reserve control and handoff resources; (6) test the venue before gates open; and (7) remove or power down temporary capacity after the event.

Effect on SIR: shorter microcell links increase desired received power, while sector antennas and coordinated reuse reduce dominant interference. However, 33 tightly spaced cell-equivalents can create substantial co-channel interference if frequency groups are assigned carelessly. The 18 dB threshold must be verified using the actual temporary-site coordinates and antenna patterns. Effect on cost: temporary equipment, transport, installation, backhaul, power, event staffing and pre/post-event optimisation are added for four hours of service; the operator avoids the recurring cost of permanently maintaining a dense 33-cell stadium layer.

12D. Evidence-Based Defence (2 Marks)

The temporary modification is the most defensible choice. The event load is 1,250 Erlangs, while the two-cell normal-layout estimate supports only about 76.78 Erlangs at the required 2% blocking. The overload factor is approximately 16.28. Retaining the design unchanged would cause severe call blocking. Permanently redesigning the whole township would be disproportionate because the event is predictable, lasts only four hours and occurs about twice per month. Temporary microcells/COWs concentrate additional capacity in the stadium zone only when needed.

The 33 cell-equivalent estimate is derived directly from the 2%-blocking Erlang-B capacity of 48 channels per cell. It is a lower-bound planning figure because it assumes balanced traffic, ideal channel reuse, no outages and no additional reserve. The final event plan should include headroom, likely requiring more than 33 equivalents if the actual crowd distribution is uneven or if some channels are reserved for control and handoff. The operator should validate the event plan with a venue walk test, crowd-density map, pre-event simulation and a live operations team.

12.5 Final decision summary

Decision question	Final answer
Retain, modify or redesign?	Modify temporarily for event days; retain normal macro design outside the event.
Why not retain unchanged?	1,250 E event demand is about 16.28 times the two-cell normal-layout 2%-blocking capacity.
Why not permanent full redesign?	The surge is local, predictable, four hours long and occurs about twice monthly.
Temporary capacity estimate	At least 33 cell-equivalents at 48 channels each; approximately 11 three-sector nodes as an idealised starting estimate.
Temporary radius	Approximately 108 m per ideal cell-equivalent in the 1 km² zone.
Required checks	SIR ≥18 dB, blocking ≤2%, coordinated channel plan, backhaul, handoffs and venue testing.